Dr.Listner: project proposal, version 2

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Summary

Dr.Listner is a visit-gap checker for older adults with hearing loss and low health literacy, starting with outpatient chest pain. It stays quiet during the visit, then flags what the visit missed and gives the patient a plain-language plan. It was built for IEEE Hack for Humanity, Track 2: AI for Accessibility and Disability Inclusion.

AI scribes record what was said. Dr.Listner checks what was never asked, never answered, or never understood, and every flag cites the transcript line or chart entry behind it.

Version 2 adds a fourth gap type (comprehension breakdowns), turn-by-turn processing, patient profiles, and a retrieval-grounded patient summary. As of the repo today, the engine through Phase 3 is built and tested. Detector quality has not yet been measured, and the eval harness, UI and deploy are still open with a deadline of Oct 21, 2026.

Problem and evidence

Roughly 1 in 20 US adults is misdiagnosed in outpatient care each year, and the patients least able to catch a gap in the visit are the most exposed to it. Figures below are carried over from the v1 report; the BMJ, JAGS, PMC hearing and JMIR items were taken from search excerpts only and should be opened before submission.

Finding	Figure	Source
Outpatient diagnostic error	About 12 million US adults a year, around half potentially harmful	BMJ Quality & Safety, 2014
Diagnostic error rate	5% of US adults seeking outpatient care each year; poor communication among clinicians, patients and families is a contributor	National Academies, 2015

Finding	Figure	Source
Hearing loss in older adults	Disabling loss in 22% of adults 65 to 74 and 55% of those 75 and older; about 30% of adults 70+ who could benefit from hearing aids have used them	NIDCD, NIH
Hearing loss in communication research	Fewer than a quarter of studies on physician communication with older patients mention hearing loss	Cohen et al., JAGS
Hearing loss and rated communication	Lower rated communication quality in a nationally representative US sample	PMC study
Health literacy	12% of US adults proficient, 36% basic or below basic (2003 data, the latest found)	NCES, 2006
Forgetting the plan	40% to 80% of medical information forgotten immediately; almost half of what is remembered is incorrect	Kessels, J R Soc Med, 2003

The gap in the market is the checking step. Ambient scribes document the conversation, and omissions were the top error in a 356-note audit (64 notes, 18%) (JMIR Med Inform, 2026). That audit covers things said but dropped from the note, which is a different problem from things that should have been asked, which is what Dr.Listner targets. This compares the category as described in published audits; individual vendors were not audited.

What changed in version 2

Version 2 moves from a one-shot, after-the-visit checker to a turn-by-turn engine focused on one patient group. Where v2 and the v1 report disagree, v2 wins (recorded in SPEC.md section 8).

Area	Version 1	Version 2
Timing	Results after the visit ends	Visit processed turn by turn through one `process_turn` engine; batch mode is a fold over all turns
Patient focus	General chest-pain patient	Older adults with hearing loss and low health literacy, caregiver optional, set by a `PatientProfile` config
Gap types	Three	Four: adds comprehension breakdown not repaired
Speakers	Clinician, patient	Clinician, patient, caregiver; turns carry optional start and end times
Patient summary	Generated from the transcript	Grounded in retrieved patient-education text with a citation on every explanation
Structure	Single checklist	Swappable condition pack (checklist, chart schema, corpus); only chest pain is built
Input	Synthetic JSON transcripts	One `TurnSource` interface: replay now, typed and microphone later
Live demo	Stretch goal	Phase 7 role-play demo, only after Phases 1 to 6 pass; typed and mic sources are out of scope for the Oct 21 deadline

Solution

Dr.Listner reads the visit against a published chest-pain checklist and the patient's chart, and raises four kinds of gap. It documents completeness only; it never suggests a diagnosis or treatment, and a clinician decides what to do with every flag.

Gap	What it catches	When it surfaces
1. Red-flag question not asked	A checklist item with no matching question	Wrap-up only, since the item may still be asked later

Gap	What it catches	When it surfaces
2. Patient concern unanswered	A worry the patient voiced that the clinician never addressed	Wrap-up only
3. Statement contradicts the chart	Patient says no heart history while the chart lists a prior stent	Immediately, as a passive badge
4. Comprehension breakdown not repaired	Patient did not hear or understand and the clinician did not repair it	Passive badge during the visit; unrepaired items at wrap-up

Gap 4 cues are our own draft and are not clinically validated: a repeat request such as "what?", an answer that does not match the question just asked, a bare "okay" after a multi-part instruction, and a plan, medication or warning-sign instruction with no teach-back. A repair is a repeat, rephrase, confirmation or teach-back.

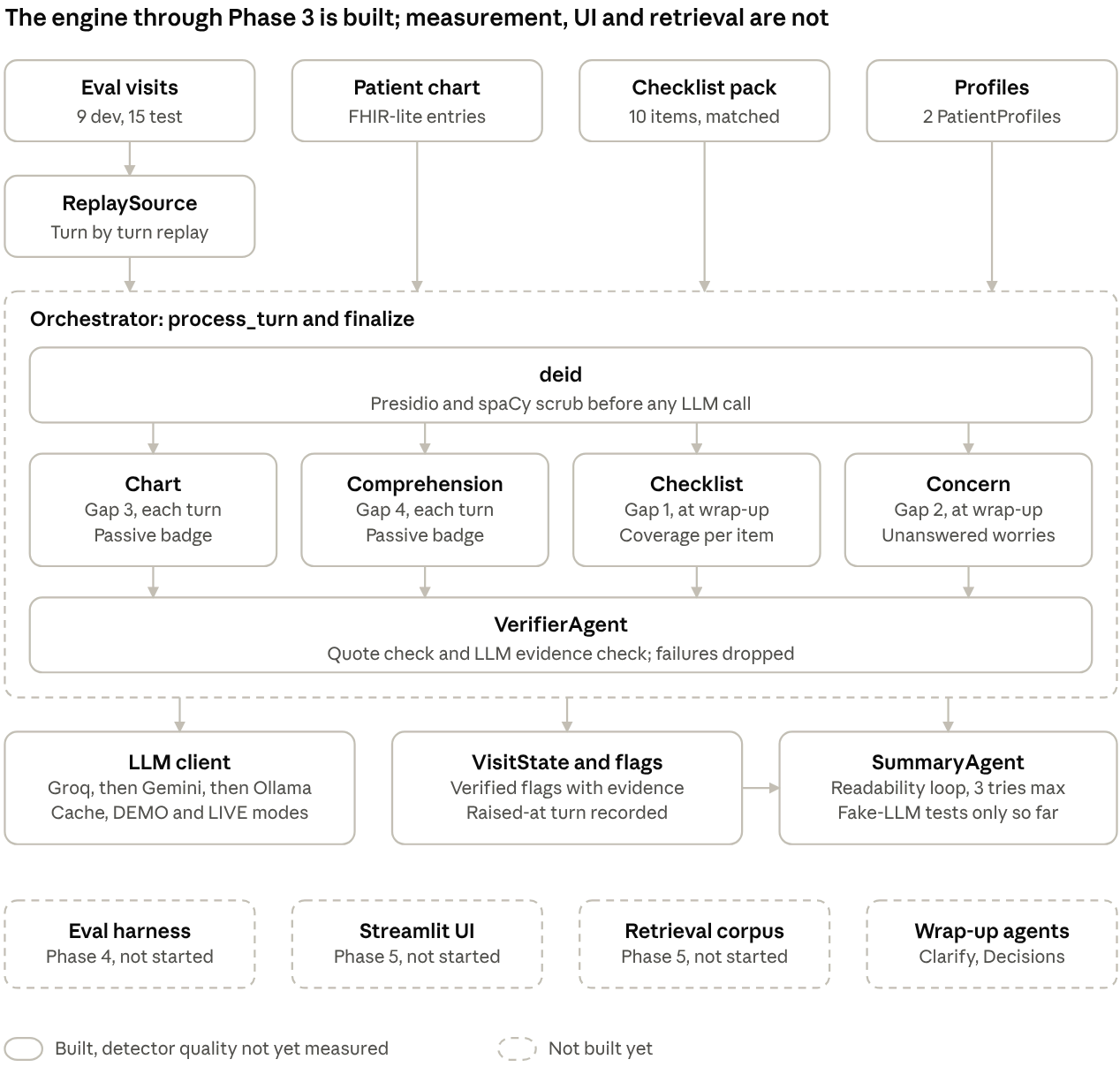
The checklist has 10 items: onset and duration, nature, relieving factors, exertional pattern, radiation, sweating, shortness of breath, fainting, family history and risk factors. All 10 were matched to the 2021 AHA/ACC chest pain guideline (Section 2.1 and Table 3) by Claude and are not clinician-reviewed.

Two outputs

- Clinician view: live coverage ticks, passive badges, a wrap-up check and an optional clarify card of at most 3 items. Each flag shows its evidence line and has a one-tap dismiss, and dismissals are logged for the eval.
- Patient view: a plain-language summary at or below a 6th-grade reading level (scored with textstat), warning signs, questions for next time, captions and read-aloud. Explanations are grounded in retrieved patient-education text with citations.

Alert-fatigue guard. During the visit there are no pop-ups, sounds or interruptions, only checklist ticks and at most passive badges for gaps 3 and 4.

Architecture



current architecture · built versus not built

Inputs feed a deterministic orchestrator that de-identifies each turn, runs four narrow gap agents and passes every candidate flag through a verifier. Chart and Comprehension checks run per turn; Checklist and Concern run once at wrap-up, so gaps 1 and 2 can never fire early. Dashed boxes are in the v2 plan but not yet in the repo.

Build status today

Phases 1 to 3 are implemented in the repo, but no detector accuracy has been measured and no UI exists. This table comes from the repo's SPEC.md and PROGRESS.md; the source folders themselves could not be read, so it is a claim from the docs, not a code audit.

Piece	Status	Note
Schemas, SPEC, config	Done	Pydantic schemas; 24 tests passed at end of Phase 1
Checklist pack	Done	10 of 10 items matched to the guideline; not clinician-reviewed
LLM client	Done	Cache, retry, DEMO and LIVE; default order Groq `gpt-oss-120b`, then Gemini, then Ollama
Provider checks	Partial	Groq verified live; Gemini and Ollama tested only against mocked HTTP
Synthetic dataset, fixtures	Done	29 template visits, dev-only contract fixtures, not for reported metrics
Eval dataset	Done, awaiting spot-read gate	9 dev and 15 test visits, 7 clean, all four gap types, caregiver visits, hard negatives
Test gap counts	Small	Type 1: 6, type 2: 5, type 3: 5, type 4: 6
De-identification	Done	Presidio with spaCy `en_core_web_sm` and a clinical keep-list; best effort
Agents and engine	Done, unmeasured	Chart, Comprehension, Checklist, Concern, Verifier, Summary; 62 tests pass against a scripted fake LLM
Live smoke test	Not a measurement	Dev only, 2 visits: both planted gap-4 cases found plus 1 extra flag; the clean visit still gave 3 false flags
Summary agent	Untested live	Fake-LLM tests only; plan coverage unmeasured

Piece	Status	Note
Eval harness, UI, retrieval, deploy	Not started	Phases 4 to 6

Known weaknesses in the data: charts are hand-authored, not Synthea; phrasing comes from a limited pool; there is no clinician review; and the team authored both sides of every gap-4 case, so gap-4 numbers measure cue detection, not real hearing-loss speech.

Tech stack

Python end to end, free or open-source pieces only, sized so a two-person team can demo within a day. The stack is the v1 plan updated for v2 and for what the repo actually uses.

Layer	Pick	Why
UI	Streamlit	Clinician and patient views in one Python app, no frontend build
Engine	Deterministic Python orchestrator with `process_turn` and `finalize`	Same code for replay and batch; not an LLM, so behavior is testable
Schemas	Pydantic	Validates every model response before it reaches the UI
LLM	Groq `gpt-oss-120b` first, then Gemini free tier, then Ollama; JSON mode, on-disk cache, DEMO mode	Free tiers, offline fallback, zero-API demo replays
Matching	Hardcoded verified checklist, LLM check at runtime	Hosted app has no PyTorch; embeddings are offline only
De-identification	Presidio with spaCy `en_core_web_sm`	Scrubs text before any hosted call
Synthetic charts	FHIR-lite view, Synthea importer	No real patient data; included charts are hand-authored so far

Layer	Pick	Why
Summary retrieval	BM25 (`rank_bm25`) over MedlinePlus, CDC and AHA patient pages	Cheap, no PyTorch; every explanation carries a source URL
Readability	textstat	Automatic grade score on every summary
Read-aloud and captions	Browser SpeechSynthesis and Web Speech API	No install
Speech to text (later)	Web Speech API or `faster-whisper`	Out of scope for the Oct 21 deadline
Evaluation	pandas, scikit-learn	Precision, recall and F1 per gap type
Hosting	Streamlit Community Cloud, GitHub	Free public demo link

Data and evaluation plan

The proof is a number: how many planted gaps Dr.Listner finds and how many false flags it raises per visit. No metric appears in any document unless it comes from a saved run under `eval/results/`.

Data. Each visit starts as a complete, gap-free script, then gaps are planted by deterministic edits: delete a red-flag question and answer, insert an unanswered worry, make a statement contradict the chart, insert a mishearing with no repair, or remove a teach-back. The answer key comes from the edit, never from a model's opinion. The split is 9 dev and 15 test visits, 7 of them clean negative controls, with dev and test phrasing kept disjoint.

Matching rule. One flag matches at most one planted gap, greedily and one to one. Type 1 matches on the checklist item id. Types 2 and 4 match when a flag's transcript pointer falls in the gap's turn indices. Type 3 matches on the chart entry or the transcript turn.

Metrics. True positive is a flag matched to a planted gap; false positive is any other flag, including any flag on a clean visit; false negative is a planted gap with no flag. Report precision, recall and F1 per gap type and overall, plus false flags per visit, flags raised too

early (type 1 or 2 before the wrap-up turn), and turns to detection for types 3 and 4. Counts are always shown beside percentages because test counts are small.

Baseline. An LLM writes a scribe-style note, then one prompt lists anything missing, with no checklist, chart agent or verifier. Its flags are scored with the same rule.

Draft targets, not promises: recall at least 80%, precision at least 80%, at most 1 false flag per visit, summaries at or below a 6th-grade reading level, and every flag citing its evidence. Gap 4 may land below target; the plan is to report that honestly with failure patterns.

Discipline. Tune prompts on the dev split only. Run the test split once per milestone and report it as is, including bad results. Verified flags only are counted, so an LLM outage lowers recall instead of inflating precision.

Privacy, safety and honest limits

The prototype never touches real patient data, is not a medical device, and makes no HIPAA claim.

- Synthetic and public data only; `synthetic: true` is required on every transcript and chart. The app stores no transcripts.
- De-identification runs before any text reaches a hosted model. It is best effort: the small language model sometimes tags capitalized words as names, and a keep-list protects drug names.
- The tool checks documentation completeness. No flag text may suggest a diagnosis or treatment; a lint rejects flags that do. Neutral wording is required, for example "Chart lists prior stent (2022); patient said no heart history".
- Every flag is dismissible, and dismissals are logged.
- Free-tier model providers may use submitted content, so only synthetic, de-identified text is ever sent.

Limits stated up front. One visit type (outpatient chest pain). Synthetic data with hand-authored charts and a limited phrase pool. The gap 4 cues are our own draft, and the team wrote both sides of every mishearing. The checklist is matched to the guideline but not clinician-reviewed. No clinician validation unless one can be secured; if it is, the results slide should say what they changed. About 25 visits give wide intervals on per-type metrics.

Deployment and cost

Target spend is $0: the app runs on Streamlit Community Cloud from a GitHub repo, model calls use free tiers, and every other component is open source. Free-tier limits change, so check them before the build.

Piece	Where it runs	Watch-out
Web app	Streamlit Community Cloud, deployed from GitHub	Docs list 690 MB RAM minimum and 2.7 GB maximum (February 2024 figures); apps idle for 12 hours sleep and wake with one click (Streamlit docs)
LLM calls	Groq, Gemini free tier, Ollama fallback	Groq's free tier is 8,000 tokens per minute and a full visit takes about 20 to 30 calls, so a first live pass is slow; cached reruns are instant
De-id, readability, eval	Presidio, spaCy, textstat, scikit-learn in the app process	The spaCy model is not yet in `requirements.txt`; footprint under 690 MB is unchecked
Synthetic data and cache	Generated offline, committed to the repo as JSON	Keep files small; the committed cache holds model outputs on synthetic text only
Backup demo	Laptop with Ollama and the same code, plus a screen recording	Covers wifi, rate-limit and sleeping-app failures

Hugging Face Spaces is not the plan, because its docs say Gradio and Docker Spaces need a paid plan to create (Hugging Face docs).

Deploy steps: keep one repo; add a Demo mode that reads cached results with no API calls and a Live mode that calls the models; store API keys in the platform's secrets settings, never in the repo; connect the repo in Streamlit Community Cloud; and within 12 hours before presenting, wake the app, run one visit end to end, and keep the recording ready.

Phases, gates and the cut line

The deadline is Oct 21, 2026 and Phases 4 to 6 are open, so the cut line decides what ships. Each phase ends at a gate that needs the team's approval before the next begins.

Phase	Scope	Gate	Status
1. Foundation and spec	Schemas, SPEC, checklist pack, profiles, LLM client	Schemas validate; checklist items cited or marked unverified	Done
2. Synthetic data and key	Charts, visits, gaps planted by edit, keys from the edits	Transcripts validate; spot-read of 5 including 2 gap-4 cases	Built; awaiting the spot-read
3. Core engine	De-id, four gap agents, verifier, orchestrator	Turn-by-turn replay gives schema-valid flags with real evidence; no omission before wrap-up	Implemented; quality unmeasured
4. Eval harness	Matching, per-type metrics, baseline, results as JSON and CSV	One command reproduces the eval; test numbers saved; top-5 error patterns	Not started
5. Retrieval summary and UI	Corpus, retriever, summary loop, Streamlit views, Play and Pause, DEMO and LIVE	Full replay works offline in DEMO mode; every explanation has a source	Not started
6. Harden and submit	Deploy, memory check, backup path, recording, README, results slide	Public link works from a fresh browser; backup works with wifi off	Not started
7. Live role-play	Typed then mic input behind the same `TurnSource`	3-minute role-play at both levels; clean card gives at most 1 flag	Out of scope for Oct 21

Cut line, dropping from the bottom if time runs short:

1. Phases 1 to 4 with saved eval numbers.

2. Phase 5 summary and UI in DEMO mode.
3. Phase 6 deploy and recording.
4. Phase 7 typed role-play, then mic.

The eval numbers and the replay demo are never traded away for the live demo. Done means a reproducible eval command, a working replay demo, a deployed link, a backup recording, a README with honest limits, and a results slide sourced from `eval/results/`.

Suggested order for the remaining time (a proposal, not in the repo): clear the Phase 2 spot-read, build the Phase 4 harness and tune on dev, run the test split once, then the minimal DEMO-mode UI, then the deploy and recording.

Production roadmap

Everything in this section is a proposal for a real deployment, not a claim about the prototype and not part of the Oct 21 scope. The report already commits to a HIPAA-eligible path as a roadmap item: a signed business associate agreement, audit logging and patient consent for recording.

Addition	What it does	Main risk
Consent-gated audio capture	Records only after patient consent; audio is not stored	Consent flow and clinic workflow fit
Streaming speech to text with speaker labels	Replaces the manual speaker toggle with automatic clinician, patient and caregiver turns	Automatic speaker labeling is the hard part and may degrade accuracy
EHR connector	Reads the real chart over read-only FHIR instead of hand-authored files	Integration effort per hospital
HIPAA-eligible cloud with a signed BAA	Replaces free tiers and Streamlit Community Cloud for real patient data	Cost and compliance work

Addition	What it does	Main risk
Clinician-signed pack registry	Each checklist version is reviewed and signed before release; a new condition is a new pack, not an engine change	Clinician time
Audit log	Records who saw which flag and when, plus consent	Retention and access policy
Dismissal feedback and monitoring	Rejected flags and per-gap precision feed tuning and drift alerts	Alert fatigue returns if unmonitored

The core design stays unchanged in production: a deterministic orchestrator, narrow agents, a verifier on every flag, a quiet visit, and wrap-up as the only time gaps 1 and 2 surface.

Risks and open items

Risk or open item	Why it matters	Mitigation
Two days to deadline with Phases 4 to 6 open	Eval numbers are the one thing the cut line protects	Build the harness first; drop live role-play
False flags on clean visits	A smoke run still gave 3 flags on a clean visit; symptom answers read as concerns	Tune on dev with a scored harness; verifier is the main control
Gap 4 may miss target	Cues are a draft and both sides of each case are team-written	Report honestly with failure patterns
Rate limits	Groq free tier is 8,000 tokens per minute; a visit takes 20 to 30 calls	Cache results and use DEMO mode for the demo
Unverified providers	Gemini and Ollama were only run against mocked HTTP; the default Gemini model name is unverified	Run the smoke script with a key and with Ollama installed

Risk or open item	Why it matters	Mitigation
Hosted footprint	spaCy model not in `requirements.txt`; memory under 690 MB unchecked	Check in Phase 6; keep PyTorch out of the hosted app
Phase 2 gate	The user spot-read of 5 transcripts is still pending	Suggested reads: eval-test-14, eval-test-10, eval-dev-08, eval-test-03, eval-test-12
Weak data variety	Hand-authored charts, templated phrasing, no clinician review	State in the README; run Synthea import if time allows
Unopened sources	BMJ, JAGS, PMC hearing study and JMIR audit were seen only as search excerpts	Open them before submitting
Guideline item check	The checklist was matched by Claude, not a clinician	Say so on the results slide; seek a clinician review if possible

Sources

Opened and checked in the v1 report: NIDCD, NCES, Kessels, National Academies brief, Google AI pricing, Streamlit docs, Hugging Face docs. Taken from search excerpts only: BMJ, JAGS, the PMC hearing study, the JMIR audit. Repo status comes from the pages listed last.

- Frequency of diagnostic errors in outpatient care, BMJ Quality & Safety (2014)
- Improving Diagnosis in Health Care, National Academies (2015)
- Quick Statistics About Hearing, NIDCD, NIH
- Hearing loss in physician-patient communication research, JAGS review
- Hearing loss and ratings of patient-physician communication, PMC
- The Health Literacy of America's Adults, NCES 2006-483
- Patients' memory for medical information, Kessels (2003)
- AI scribe quality audit, JMIR Medical Informatics (2026)
- 2021 AHA/ACC Guideline for the Evaluation and Diagnosis of Chest Pain
- Streamlit Community Cloud docs

- Spaces overview, Hugging Face docs
- Dr.Listner repository, including SPEC.md and PROGRESS.md